

MS1411 – MATHEMATICAL STATISTICS

WEEK: 4

LECTURE TOPICS:

- NORMAL DISTRIBUTION
- NORMAL APPROXIMATION TO THE BINOMIAL
- CHEBYSHEV'S RULE TRANSFORMATION OF VARIABLES
- WEIBULL DISTRIBUTION AND RELIABILITY
- ESTIMATION AND MAXIMUM LIKELIHOOD ESTIMATION

REFERENCE TEXT: WALPOLE ET AL., PROBABILITY & STATISTICS FOR
ENGINEERS & SCIENTISTS, 9TH EDITION, PEARSON

AGENDA

Normal Distribution Remainder

- Properties & Characteristics

Normal Approximation to the Binomial

- Conditions for approximation
- Continuity correction

Chebyshev's Rule

- Statement
- Practical interpretation

Estimating Probability within a Given Standard Deviation

- Connections to Chebyshev, Empirical Rule

Transformation of Variables

- Monotonic transformations
- Density transformations

Weibull Distribution and Reliability

- PDF, CDF
- Failure rate, mean time to failure

Estimation

- Point estimators
- Bias, mean squared error

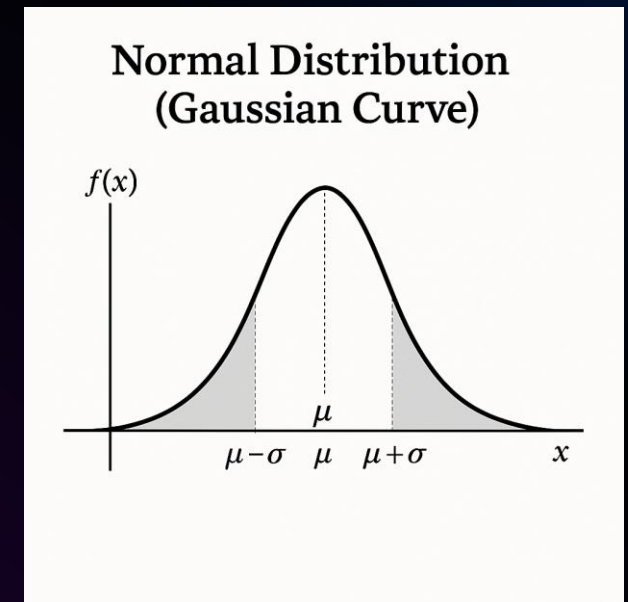
Maximum Likelihood Estimation (MLE)

- General principles
- Examples with the Normal and Weibull

NORMAL DISTRIBUTION

A random variable X is **normally distributed** with mean μ and variance σ^2 if

- $X \sim N(\mu, \sigma^2) \Rightarrow f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty.$
- **Shape:** Bell-shaped, symmetric about μ .
- Properties:
 - Symmetric about μ (mean).
 - σ controls spread (standard deviation).
 - Inflection points at $\mu \pm \sigma$.



NORMAL APPROXIMATION TO THE BINOMIAL

- Why Do We Approximate the Binomial with the Normal?
 - **Binomial Setting:** A random variable X follows a Binomial distribution, $X \sim \text{Binomial}(n, p)$, when:
 - You have a fixed number of independent trials n .
 - Each trial has only two possible outcomes (often called “success” or “failure”).
 - The probability of success p remains the same from trial to trial.
 - **Motivation for Approximation:**
 - Large n : For large sample sizes, computing exact binomial probabilities can be difficult.
 - Normal-Like Shape: By the Central Limit Theorem, the Binomial distribution often resembles a bell-shaped curve for large n .
- Conditions for a “Reasonable” Normal Approximation
 - Rule of Thumb: Both np and $n(1-p)$ should be at least 5 (some texts say 10). This ensures the binomial shape is not too skewed.
 - Mean and Variance of X :
 - Mean: $\mu = np$
 - Variance: $\sigma^2 = np(1-p)$
 - Standard deviation: $\sigma = \sqrt{np(1-p)}$.
 - Hence, the approximate distribution is:
 $X \approx Y \sim N(np, np(1-p))$.

NORMAL APPROXIMATION TO THE BINOMIAL

- Example:
 - A survey shows that 40% of electronic components are defective. Suppose you select $n=100$ items. Then X , the number of defective items, follows a binomial with $p=0.40$.
 - Step 1: Define parameters
 - $\mu = np = 100 * 0.40 = 40, \sigma = \sqrt{np(1-p)} = \sqrt{100 * 0.4 * 0.6} = \sqrt{24} = 4.9$
 - Step 2: Approximation
 - $X \approx N(40, 4.9^2)$
 - Step 3: Continuity Correction (e.g. Probability of number of defectives higher than 45 $P(X \geq 45)$)
 - $P(X \geq 45) \approx P(Y \geq 44.5)$ where $Y \sim N(40, 4.9^2)$
 - Step 4: Standardize for easier calculation:
 - $Z = \frac{Y-\mu}{\sigma} = \frac{Y-40}{4.9}, P(Y \geq 44.5) = P(Z \geq \frac{44.5-40}{4.9})$
 - Use the standard normal table or software to compute this probability.
- Continuity Correction
 - Problem: The Binomial variable X takes integer values, while the normal distribution Y is continuous. This mismatch can reduce the accuracy of the approximation.
 - Solution: Continuity correction:
 - When you calculate a probability of the form $P(a \leq X \leq b)$, replace it with $P(a-0.5 < Y < b+0.5)$, where Y is the normal approximation with $\mu=np$ and $\sigma=\sqrt{np(1-p)}$.

CHEBYSHEV'S RULE

Chapter 4.4 (Walpole et al.)

- The Statement of Chebyshev's Inequality
 - For Any Distribution (regardless of shape), if a random variable X has mean μ and standard deviation σ , then for any $k > 1$,
 - $P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$
 - This gives a lower bound on the proportion of values (or probability) that lie within k standard deviations of the mean.

Key Points

- **Works for Any Distribution:**
 - No assumption of normality or symmetry.
 - Applies even if the distribution is skewed or has outliers.
- **Guaranteed Bounds:**
 - Tells us “at least how many” observations are in the interval $\mu \pm k\sigma$.
 - Unlike the empirical (68–95–99.7) rule, which is specific to normal distributions, Chebyshev's rule is more general but often less precise.
- **Values of k :**
 - If $k=2$: At least $1 - 1/4 = 75\%$ of observations lie within 2σ of the mean.
 - If $k=3$: At least $1 - 1/9 = 88.9\%$ of observations lie within 3σ .
 - If $k=4$: At least $1 - 1/16 = 93.75\%$ lie within 4σ .

CHEBYSHEV'S RULE

Why Use Chebyshev's Rule?

- **Non-Normal or Unknown Shape:** If you cannot assume the data follow a bell curve, Chebyshev's gives a guaranteed lower bound.
- **Large Outliers:** Chebyshev's inequality still works even if the distribution has very heavy tails.
- **Broad Applicability:** In reliability, finance, or any field where distribution shape is unknown, it sets a universal minimum on how the data cluster around μ .

Examples

1. **Non-Symmetric Data:** Suppose a set of measurements is highly skewed—some extremely large values, some very small. We might guess $\mu \approx 50$ and $\sigma \approx 10$, but not assume normality. Then Chebyshev's tells us that:
 1. $P(|X - 50| < 20) \geq 1 - \frac{1}{4} = 0.75$
 2. In other words, *at least* 75% of the observations lie between 30 and 70. (2σ of the mean)
2. **General Probability Bound:** If your data or random variable are unknown or complicated in shape, and someone says “What fraction is within 3 standard deviations?” Chebyshev's ensures it is *at least* 88.9%. The actual fraction could be higher, especially if the data are near normal, but cannot be lower.

ESTIMATING PROBABILITY WITHIN A GIVEN STANDARD DEVIATION

1. **Goal:** Estimate $P(|X - \mu| \leq k\sigma)$ for a random variable X .

2. **Approaches:**

- If X is normal: Use the standard normal cumulative tables or software.
- If X is unknown: Chebyshev's inequality provides a guaranteed lower bound.

Example A: Normal Distribution Case

- Context: Suppose a machine cuts metal rods for an assembly process. The length X (in cm) of each rod is approximately normally distributed with mean $\mu=50$ and standard deviation $\sigma=0.8$.
- Question: What is the probability that a randomly selected rod is within 1.5 cm of the mean (i.e., between 48.5 cm and 51.5 cm)? Formally,
- $P(48.5 \leq X \leq 51.5) = P(|X - 50| \leq 1.5)$.
- Method:**
 - Convert to the standard normal variable $Z = (X - \mu)/\sigma$.
 - Then $48.5 \leftrightarrow Z = \frac{48.5 - 50}{0.8} = -1.875$ and $51.5 \leftrightarrow Z = \frac{51.5 - 50}{0.8} = +1.875$.
 - So we need $P(-1.875 \leq Z \leq 1.875)$.
- Calculation:
 - Use a standard normal table or software.
 - Typically, $P(Z \leq 1.875) \approx 0.9699$, and $P(Z \leq -1.875) \approx 1 - 0.9699 = 0.0301$ (by symmetry it should be about 0.0301).
 - Hence, $P(-1.875 \leq Z \leq 1.875) = 0.9699 - 0.0301 = 0.9398$.
 - So about **93.98%** of rods lie within 1.5 cm of the mean.

ESTIMATING PROBABILITY WITHIN A GIVEN STANDARD DEVIATION

1. Example B: Large-Sample/Non-Normal Case with Chebyshev

- a. Context: Suppose we are studying the daily yield Y from a chemical process. We have an estimated mean $\mu=1000$ liters and standard deviation $\sigma=50$ liters from historical data. But the shape of Y is not known (it might be skewed or irregular).
- b. Goal: We want at least a guaranteed minimum probability that the daily yield is within 2 standard deviations of its mean—i.e., in the interval $[900, 1100]$.
- c. **Apply Chebyshev:** $P(|Y - 1000| \leq 2 * 50) \geq 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = 0.75$.
- d. Interpretation:
 - 1) Chebyshev tells us: At least 75% of the time, the yield is between 900 and 1100 liters.
 - 2) The actual fraction could be higher. But Chebyshev's inequality ensures it can't be lower.
 - 3) If the process is suspected to be near normal, we could guess it's probably closer to 95% in that interval (per the “empirical rule”).
 - 4) But Chebyshev's result is a “worst-case, guaranteed **lower bound**” that works regardless of distribution shape.

TRANSFORMATION OF VARIABLES

Why Do We Transform Variables?

- Re-parameterization: We often define a new variable $Y=g(X)$ for convenience or theoretical reasons (e.g., linearizing data, simplifying integrals).
- Distribution Discovery: If the distribution of X is known, we can find the PDF (or PMF) of Y .
- Engineering & Science: Converting physical measurements (e.g., Celsius to Fahrenheit or log-transforming data) is extremely common.

Monotonic Transformations

- If $Y=g(X)$ is a strictly monotonic function (only increasing or only decreasing) and differentiable, then for each y there is exactly one $x = g^{-1}(y)$. The PDF of Y can be derived via the chain rule:
- Start: X has PDF $f_X(x)$.
- Invert: $x = g^{-1}(y)$.
- Derivative of the Inverse: $\frac{dx}{dy} = \frac{d}{dy} g^{-1}(y)$.
- Result (PDF of Y): $f_Y(y) = f_X(g^{-1}(y)) \times \left| \frac{d}{dy} g^{-1}(y) \right|$
- Increasing vs. Decreasing:
 - If g is increasing, the absolute value of the derivative is straightforward.
 - If g is decreasing, the sign changes, but the formula still has an absolute value.

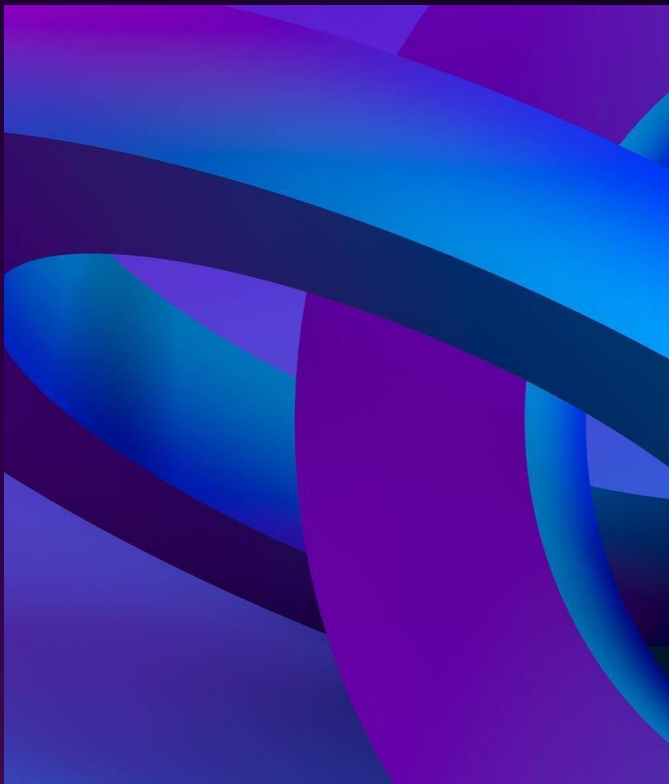
Example: $Y=a+bX$

- If $Y=a+bX$ with $b \neq 0$ and X has PDF $f_X(x)$, then
 - $x = \frac{Y-a}{b}, \frac{dx}{dy} = \frac{1}{b}$.
- Then:
 - $f_Y(y) = f_X\left(\frac{y-a}{b}\right) \left| \frac{1}{b} \right|$
- Interpretation: If X is normal(μ, σ^2), then $Y=a+bX$ is also normal($a+b\mu, b^2\sigma^2$).

Log Transformation Example

- Suppose $X>0$ has PDF $f_X(x)$. Define $Y=\ln(X)$. Then $x = e^y$, and $\frac{dx}{dy} = e^y$
- Therefore: $f_Y(y) = f_X(e^y) |e^y| = f_X(e^y) e^y$, valid for $-\infty < y < \infty$.
- Common Usage: If X is lognormal, then $Y=\ln(X)$ is normal.

TRANSFORMATION OF VARIABLES



Piecewise Monotonic Transformations

If $g(X)$ is not strictly monotonic over the entire domain but can be broken into segments where it is monotonic, handle each segment separately and then add up the results. For instance, if g is monotonic on intervals (a,c) and (c,b) but changes direction at $x=c$, the PDF for Y is a sum of the PDFs obtained for each region:

$$f_Y(y) = \sum_{j=1}^m [f_X(g_j^{-1}(y)) \left| \frac{d}{dy} g_j^{-1}(y) \right|]$$

Where g_j denotes the piecewise function on the j -th monotonic interval

Clarification: A function $g(x)$ is called monotonic on a certain interval if it never reverses its direction there, that is, it is either consistently non-decreasing (always going up or staying level) or consistently non-increasing (always going down or staying level) throughout that interval.

TRANSFORMATION OF VARIABLES

1. Practical Tips

- a. Check Monotonicity: Confirm g is 1-to-1 on the domain of X . If not, partition the domain.
- b. Use Absolute Value: Always include $|dx/dy|$.
- c. Respect Domain: Make sure to incorporate constraints where $f_X(x)$ is zero (e.g., if $X > 0$, do not allow negative values in the new PDF).
- d. Common Transformations:
 - 1) Linear: $Y = a + bX$.
 - 2) Log: $Y = \ln(X)$.
 - 3) Power: $Y = X^\beta$ or $Y = X^\beta$.
 - 4) Reciprocal: $Y = 1/X$.

WEIBULL DISTRIBUTION AND RELIABILITY

1. What Is the Weibull Distribution?

- Definition: A random variable X has a Weibull distribution if its PDF (probability density function) is given by
 - $f(x; \alpha, \beta) = \alpha \beta x^{\beta-1} e^{-\alpha x^\beta}, x > 0, \alpha > 0, \beta > 0.$
 - Here:
 - β is called the shape parameter.
 - α is the scale parameter.
- CDF
 - $F(x) = 1 - e^{-\alpha x^\beta}, x > 0.$
- Interpretation
 - The Weibull is frequently used in engineering contexts for time to failure data (e.g., lifetimes of machines, bearings, or electronic components).

• Important Characteristics

- Mean (Expected Value)
 - $\mu = \alpha^{-\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right),$ where Γ is the gamma function
- Variance
 - $\sigma^2 = \alpha^{-\frac{2}{\beta}} \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \Gamma\left(1 + \frac{1}{\beta}\right)^2 \right].$
- Relation to the Exponential
 - If $\beta=1$, then X becomes an exponential distribution with parameter α . In that case,
 - $f(x) = \alpha e^{-\alpha x}, x > 0.$

WEIBULL DISTRIBUTION AND RELIABILITY

- Reliability Function and Failure Rate

- In reliability theory, we often focus on the probability that a system (or component) survives up to a certain time t .

- Reliability Function $R(t)$:

$$R(t) = 1 - F(t) = e^{-(\alpha * t^\beta)}$$

Interpreted as: Probability the component is still functioning (not failed) by time t .

- Hazard (Failure) Rate $\lambda(t)$:

$$\lambda(t) = \frac{f(t)}{R(t)} = \alpha \beta t^{\beta-1}$$

This gives the instantaneous failure rate at time t .

Not constant in general (unlike the exponential distribution).

- β and Failure Rate Behavior

- $\beta > 1$: $\lambda(t)$ increases over time $\Rightarrow$ wear-out mode.
- If $\beta = 1$: $\lambda(t)$ is constant $\Rightarrow$ exponential distribution (memoryless).
- If $\beta < 1$: $\lambda(t)$ decreases over time $\Rightarrow$ early “infant mortality” is more likely.

Simple Example

Context: A certain type of electrical insulation fails over time. Tests suggest $\beta=2$ (shape) and $\alpha=0.001$ (scale).

PDF:

$$f(x) = (0.001)(2)x^{2-1}e^{-0.001x^2} = 0.002x e^{-0.001x^2}, x > 0$$

Reliability for time t :

$$R(t) = e^{-0.001t^2}$$

This is the probability that the insulation is still functional at time t .

Failure Rate:

$$\lambda(t) = 0.001 \times 2t^{2-1} = 0.002t$$

Increases linearly with t . The older the insulation, the higher the chance of failure per unit time (wear-out phenomenon).

Why Is Weibull Useful in Reliability?

- Flexibility: By adjusting β , one can model a wide range of behaviors: burn-in, random failure, wear-out, etc.
- Engineers & Manufacturers often gather time-to-failure data, fit a Weibull, and then:
 - Predict lifetimes: E.g., “What fraction fails before 5000 hours?”
 - Set warranty policies: Using the distribution’s reliability.
 - Compare designs: Checking if a new design has lower α or higher β (better reliability).

ESTIMATION – BASIC CONCEPTS

Reference: *Walpole et al.*, 9th edition, Chapters 9.1–9.4

The Goal of Estimation

- We want to estimate unknown parameters (e.g., population mean μ , variance σ^2 , proportion p) using data from samples.
- Common examples:
 - The sample mean $\bar{X}$ estimates the population mean μ .
 - The sample variance S^2 estimates the population variance σ^2 .
 - The sample proportion $p \approx \frac{x}{n}$ estimates the population proportion p .

Point Estimators

- Definition: A point estimator of a parameter θ is a single-number statistic $\hat{\theta}$ computed from sample data and used to approximate θ .
- Examples
 - Mean: $\mu \approx \frac{1}{n} \sum X_i$
 - Variance: $\sigma^2 \approx S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$.
 - Success Probability: $p \approx \hat{p} = \frac{x}{n}$.

ESTIMATION – BASIC CONCEPTS

1. Important Properties of Estimators

a. Unbiasedness:

- 1) An estimator $\hat{\theta}$ is unbiased for θ if $E[\hat{\theta}] = \theta$. (If expected value of the estimator is equal to theta, the parameter it is estimating)
- 2) Example: $\bar{X}$ is an unbiased estimator of μ . The sample variance S^2 is unbiased for σ^2 .

b. Consistency:

- 1) $\hat{\theta}_n$ is consistent if $\hat{\theta}_n \rightarrow \theta$ in probability as $n \rightarrow \infty$
- 2) Intuitively, as sample size grows large, $\hat{\theta}_n$ should converge to the true parameter θ .

c. Efficiency:

- 1) Among all unbiased estimators for θ , one whose variance is smallest is called most efficient.
- 2) For large n , the sample mean $\bar{X}$ is efficient for μ

d. Sufficiency (more advanced concept):

- 1) A statistic is sufficient if it captures all needed information about θ from the sample.

ESTIMATION – BASIC CONCEPTS

1. Measuring Performance of Estimators

a. Variance $\text{Var}[\hat{\theta}]$: lower is better.

b. **Mean Squared Error (MSE):**

1)
$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2.$$

2) If $\hat{\theta}$ is unbiased $\text{MSE} = \text{Var}(\hat{\theta})$.

3) Minimizing MSE is often a primary objective

Example: Sample Mean

Suppose we have independent identical distributed (i.i.d.) observations $X_1, \dots, X_n$ from a population with mean μ .

We estimate μ via $\bar{X} = \frac{1}{n} \sum X_i$.

Unbiasedness: $E[\bar{X}] = \mu$.

Variance: $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$

As $n \rightarrow \infty$, $\bar{X}$ converges strongly to μ .

Clarification: In other words, i.i.d. means each observation comes from the same “population” or distribution, and none of the observations influences any other. This assumption underlies many standard statistical methods, ensuring that formulas for means, variances, confidence intervals, etc., are valid.

MAXIMUM LIKELIHOOD ESTIMATION (MLE)

- **Reference:** *Walpole et al.*, 9th edition, Section 9.14
- What Is the Likelihood Function?
 - Suppose $X_1, X_2, \dots, X_n$ are i.i.d. from a distribution with PMF/PDF $f(x; \theta)$, where θ is unknown
 - **Likelihood:** The *likelihood function* is
 - $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$.
 - We treat $\{x_i\}$ as fixed from the sample and view $L(\theta)$ as function of θ .
 - The goal of MLE: choose $\hat{\theta}$ that maximizes $L(\theta)$.
- Why MLE?
 - Widely Used: MLE typically provides estimators with good properties (consistency, efficiency).
 - Flexible: Applies to many distribution: normal, binomial, Poisson, Weibull, etc.
 - **Log-Likelihood:** Often easier to maximize $\ell(\theta) = \ln L(\theta)$.

MLE IN FOUR BASIC STEPS

1. Write Down $L(\theta)$: The product of the PDFs or PMFs for all observed data.
2. Take the Log: $\ell(\theta) = \ln L(\theta)$ to simplify products into sums.
3. Differentiate: $\frac{d\ell(\theta)}{d\theta} = 0$, find critical points.
4. Check the sign of the second derivative or the context to confirm maximum.

EXAMPLE MLE

- A. MLE for the Normal Mean & Variance
 - Suppose $X_i \sim N(\mu, \sigma^2)$ i.i.d. (you have sample $\{x_1, \dots, x_n\}$ from $N(\mu, \sigma^2)$)
 - **Likelihood:** $L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$.
 - **Log-Likelihood:** $\ell(\mu, \sigma^2) = -n \ln(\sigma) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$
 - **Partial Derivatives:** $\frac{d\ell}{d\mu} = 0 \Rightarrow \hat{\mu} = \bar{X}, \frac{d\ell}{d\sigma^2} = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2$.

MLE CONCLUSION

- **MLE:** A powerful, general method for parameter estimation.
- **Use Cases:** Normal, binomial, Poisson, Weibull, logistic regression, etc.
- **Connection to Next Topics:**
 - Confidence intervals (we often use MLE inside the intervals).
 - Hypothesis testing (likelihood ratio tests).

EXERCISE 1

Scenario (Poisson Defects)

- A quality-control engineer suspects that the **number of defects** on a semiconductor wafer follows a **Poisson** distribution with some unknown parameter λ . The engineer samples n wafers and observes the following defects:
- $X_1, X_2, \dots, X_n$ (each X_i is the count of defects on wafer i)
- Model: $X_i \sim \text{Poisson}(\lambda), i = 1, \dots, n$.
- Task:
 - (a) Write down the likelihood $L(\lambda)$.
 - (b) Take the natural log to form $\ell(\lambda)$.
 - (c) Differentiate $\ell(\lambda)$ with respect to λ , set equal to 0, and solve for $\hat{\lambda}$.
 - (d) Explain briefly why your result makes intuitive sense.
 - Suppose $n=6$ wafers yield defect counts $\{2, 3, 0, 1, 4, 2\}$. Compute $\hat{\lambda}$ explicitly and interpret it.

EXERCISE 1 SOLUTION

- PMF of each observation is:

$$P(X_i = x_i) = \frac{\lambda^{x_i} e^{-\lambda}}{(x_i)!}, \quad x_i = 0, 1, 2, \dots$$

- The joint distribution (likelihood) of the n sample is:

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}.$$

- If we work out a bit the expression:

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = (e^{-\lambda})^n \lambda^{\sum x_i} \prod_{i=1}^n \frac{1}{x_i!}.$$

- Form the Log-Likelihood $\ell(\lambda)$:

$$\ell(\lambda) = \ln L(\lambda) = \sum_{i=1}^n (x_i \ln \lambda - \lambda - \ln(x_i!)) = \left(\sum_{i=1}^n x_i \right) \ln(\lambda) - n\lambda - \sum_{i=1}^n \ln(x_i!).$$

- Differentiate and Set to Zero:

$$\frac{d}{d\lambda} \ell(\lambda) = \frac{\sum_{i=1}^n x_i}{\lambda} - n = 0 \implies \hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i.$$

- Computation:

- $n = 6$
- Sum of defect counts: $2 + 3 + 0 + 1 + 4 + 2 = 12$
- MLE: $\hat{\lambda} = \frac{12}{6} = 2$
- On average, you estimate 2 defects per wafer